Advanced Python Programming Lab — 03.09.2026

1. Using csv

a. Write a csv file with headers: Student Name, CGPA, Certifications_Count, Internship_Exp_yrs

b. Add 4 rows of data

In [30]:
```
import csv

h1 = ["Student Name", "CGPA", "Certifications_Count", "Internship_Exp_yrs"]
r1 = [
    ["vk", 8.7, 3, 1],
    ["vickey ", 7.9, 2, 0],
    ["Sneha ", 9.2, 5, 2],
    ["narmatha", 8.1, 1, 1],
]
```

In [36]:
```
with open("students.csv", "w", newline="") as f:
    writer = csv.writer(f)
    writer.writerow(h1)
    writer.writerows(r1)
```

h. Read the rows from the first csv file and print them all

In [28]:
```
with open("students.csv", "r", newline="") as f:
    reader = csv.reader(f)
    for row in reader:
        print(row)
```

```
['Student Name', 'CGPA', 'Certifications_Count', 'Internship_Exp_yrs']
['vk', '8.7', '3', '1']
['vickey ', '7.9', '2', '0']
['Sneha ', '9.2', '5', '2']
['narmatha', '8.1', '1', '1']
```

2. Using Pandas

a. Load student details to a Dataframe

In [38]:
```
import pandas as pd

df = pd.read_csv("students.csv")
df
```

Out[38]:

	Student Name	CGPA	Certifications_Count	Internship_Exp_yrs
0	vk	8.7	3	1
1	vickey	7.9	2	0
2	Sneha	9.2	5	2
3	narmatha	8.1	1	1

b. Use `to_csv()` to write the contents from a dataframe to a csv file

In [41]:
```
df.to_csv("students_pandas.csv")
```

c. Read from this csv and print all contents using `read_csv()`

In [42]:
```
df_read = pd.read_csv("students_pandas.csv")
print(df_read)
```

```
   Unnamed: 0 Student Name  CGPA  Certifications_Count  Internship_Exp_yrs
0           0           vk   8.7                     3                   1
1           1       vickey   7.9                     2                   0
2           2        Sneha   9.2                     5                   2
3           3     narmatha   8.1                     1                   1
```

d. Use the concatenation method to append a few more rows to the csv file

In [43]:
```
new_rows = pd.DataFrame([
    ["Priya Nair", 8.5, 4, 1],
```

```
    ["Arjun Das", 7.3, 0, 0],
], columns=["Student Name", "CGPA", "Certifications_Count", "Internship_Exp_yrs"])

df_updated = pd.concat([df_read, new_rows], ignore_index=True)
df_updated.to_csv("students_pandas.csv", index=False)
df_updated
```

Out[43]:

	Unnamed: 0	Student Name	CGPA	Certifications_Count	Internship_Exp_yrs
0	0.0	vk	8.7	3	1
1	1.0	vickey	7.9	2	0
2	2.0	Sneha	9.2	5	2
3	3.0	narmatha	8.1	1	1
4	NaN	Priya Nair	8.5	4	1
5	NaN	Arjun Das	7.3	0	0

e. Read the updated file

In [44]:
```
df_final = pd.read_csv("students_pandas.csv")
print(df_final)
```

```
   Unnamed: 0 Student Name  CGPA  Certifications_Count  Internship_Exp_yrs
0         0.0           vk   8.7                     3                   1
1         1.0       vickey   7.9                     2                   0
2         2.0        Sneha   9.2                     5                   2
3         3.0     narmatha   8.1                     1                   1
4         NaN   Priya Nair   8.5                     4                   1
5         NaN    Arjun Das   7.3                     0                   0
```

4. Using Pandas (with dictionary/JSON-style data)

a. Load a student details dictionary to a dataframe

In [ ]:
```

```

In [45]:
```
student_dict = {
    "Student Name": ["Aditi Sharma", "Rahul Verma", "Sneha Iyer", "Karan Mehta"],
    "CGPA": [8.7, 7.9, 9.2, 8.1],
    "Certifications_Count": [3, 2, 5, 1],
    "Internship_Exp_yrs": [1, 0, 2, 1],
}

df_dict = pd.DataFrame(student_dict)
df_dict
```

Out[45]:

	Student Name	CGPA	Certifications_Count	Internship_Exp_yrs
0	Aditi Sharma	8.7	3	1
1	Rahul Verma	7.9	2	0
2	Sneha Iyer	9.2	5	2
3	Karan Mehta	8.1	1	1

In [48]:
```
print(df_dict.head())
print(df.information())
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
~\AppData\Local\Temp\ipykernel_2440\64841649.py in ?()
----> 1 print(df.information())

~\anaconda3\Lib\site-packages\pandas\core\generic.py in ?(self, name)
   6317             and name not in self._accessors
   6318             and self._info_axis._can_hold_identifiers_and_holds_name(name)
   6319         ):
   6320             return self[name]
-> 6321         return object.__getattribute__(self, name)

AttributeError: 'DataFrame' object has no attribute 'information'
```

d. Inspect the data using `head()`, `info()`, `describe()` methods

In [49]:
```
print(df_dict.head())
```

```
df_dict.info()

print(df_dict.describe())
```

```
   Student Name  CGPA  Certifications_Count  Internship_Exp_yrs Placed
0  Aditi Sharma   8.7                     3                   1    Yes
1   Rahul Verma   7.9                     2                   0     No
2    Sneha Iyer   9.2                     5                   2    Yes
3   Karan Mehta   8.1                     1                   1     No
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4 entries, 0 to 3
Data columns (total 5 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Student Name          4 non-null      object
 1   CGPA                  4 non-null      float64
 2   Certifications_Count  4 non-null      int64
 3   Internship_Exp_yrs    4 non-null      int64
 4   Placed                4 non-null      object
dtypes: float64(1), int64(2), object(2)
memory usage: 292.0+ bytes
           CGPA  Certifications_Count  Internship_Exp_yrs
count  4.000000              4.000000            4.000000
mean   8.475000              2.750000            1.000000
std    0.590903              1.707825            0.816497
min    7.900000              1.000000            0.000000
25%    8.050000              1.750000            0.750000
50%    8.400000              2.500000            1.000000
75%    8.825000              3.500000            1.250000
max    9.200000              5.000000            2.000000
```

g. Add a new column and load the values

In [47]:
```
df_dict["Placed"] = ["Yes", "No", "Yes", "No"]
df_dict
```

Out[47]:

	Student Name	CGPA	Certifications_Count	Internship_Exp_yrs	Placed
0	Aditi Sharma	8.7	3	1	Yes
1	Rahul Verma	7.9	2	0	No
2	Sneha Iyer	9.2	5	2	Yes
3	Karan Mehta	8.1	1	1	No

In [ ]:

In [ ]:

In [ ]:

In [ ]: